Serotonin induces a down-state across the anesthetized mouse brain

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Abstract

Serotonin (5-HT) is a neuromodulator which is released from neurons in the raphe nucleus that project widely across the forebrain. Serotonin has been implicated in the regulation of slow oscillations which occur during sleep and light anesthesia. These so called UP and DOWN states are periods of high neural activity interspersed with complete silence, observed mostly in the cortex. fMRI studies have shown that optogenetic stimulation of serotonergic neurons in the dorsal raphe nucleus results in a brain-wide suppression of neural activity, suggesting that serotonin transitions the system into a DOWN state. These findings, however, are contradicted by electrophysiological recordings which show that electrical stimulation of the dorsal raphe nucleus induces an UP state. Here we optogenetically stimulated serotonin release while recording large parts of the forebrain with Neuropixel probes in lightly anesthetized mice. We found that 5-HT stimulation induced a DOWN state across the cortex, striatum, and amygdala, but not in the midbrain. Computational modelling revealed that serotonin likely initiated a DOWN state by inhibiting the excitatory population.

Introduction

Serotonin (5-HT) is a neuromodulator which plays a crucial role in many brain functions and modulates neural activity in various ways [1]. One of the activity patterns in the brain which is modulated by serotonin are slow oscillations, with alternating states of high and low activity, also referred to as UP and DOWN states. Slow oscillations appear during anaesthesia and slow-wave sleep and are important for learning, memory consolidation and plasticity [2], they are thought to be the default emergent activity of neural networks [2, 3]. Altough it is clear that serotonin affects slow oscillations, different studies arrive at conflicting conclusions as to the directionality of its effect.

It was shown that inceasing tonic levels of serotonin by administering MDMA supresses slow oscillations in anaesthetised mice. Furthermore, bath application of 5-HT to acute brain slices completely abolishes slow oscillations [4]. Besides these slow tonic increases in serotonin, also transient pulses of 5-HT suppress neural activity across the brain, as measured with fMRI [5, 6]. Altough these studies suggest that serotonin induces a DOWN state, they lack the temporal precision to directly establish this. Another piece of evidence which supports the notion that serotonin should induce a DOWN state is that 5-HT stimulation suppresses hippocampal sharp wave ripples [7, 8]. Since sharp wave ripples occur during UP states it is likely that serotonin transitions the neural network into a DOWN state. This is, however, only indirect evidence. In contrast to the previous findings, one study showed that electrical stimulation of the dorsal raphe nucleus (DRN) in rats induces an UP state in frontal cortex [9]. To our knowledge, this is the only study which investigated the direct effect of a transient increase of serotonin on UP-DOWN dynamics with high temporal resolution. Their findings, however, are in direct conflict with a body of literature which indicates that serotonin suppresses neural activity and transitions the network into a DOWN state. Serotonin's effect on UP-DOWN state dynamics under anesthesia is therefore still a topic for debate.

To shed light on this issue, we stimulated serotonin by optogenetically activating DRN neurons in the anesthetized mouse while recording single cell spiking activity across the brain with Neuropixel probes. We found that 5-HT stimulation rapidly induced a DOWN state in cortex, amygdala, striatum, hippocampus, thalamus, but not the midbrain. To investigate the underlaying mechanism of this effect we modelled UP-DOWN state dynamics with a firing rate model with realistic inter-area connectivity which replicates the experimental data [10, 11].

Results

We performed acute Neuropixel recordings across the mouse brain while the animal was under light anesthesia. Over four consecutive recording days, seven Neuropixel insertions were made targeting frontal cortex, visual cortex, amygdala, striatum, hippocampus, thalamus, and the midbrain. Serotonin was stimulated by activating serotonergic neurons in the DRN optogenetically during Neuropixel recordings in the forebrain (Fig. 1).

Cortical UP-DOWN states were clearly visible under anesthesia as periods of 1-2s of neural activity interspersed with periods of silence (Fig. 2a). Based on visual inspection of the spike raster plots, 5-HT stimulation appeared to coincide with a DOWN state. To investigate this, first we asked what the effect of serotonin stimulation was on the activity of single neurons. Peri-stimulus time histograms of single neurons, conditioned on the start of the 5-HT stimulation, consistently showed that single neuron spiking activity was decreased (example neuron in Fig. 1b). In all recorded brain regions we found a large fraction of neurons that were significantly modulated by serotonin stimulation (30-60%; ZETA-test [12], $p < 0.05$; Fig. 2c). A large majority of single neurons were suppressed by 5-HT stimulation as quantified by their modulation index, which was predominantly < 0. The mean modulation index over neurons per brain region was negative indicating that all recorded regions were suppressed by serotonin stimulation (Fig. 2d).

To directly investigate whether serotonin stimulation induced a DOWN state we quantified UP and DOWN states with a two-state Hidden Markov Model (HMM) which was fit to spike rates conditioned to the start of the 5-HT stimulation.

To investigate the mechanism behind the effect of serotonin stimulation in mice during anaesthesia, a dynamic firing rate model was implemented, which simulates a simplified network of the mouse brain, following biological realistic inter-area connectivity. Each brain area was modelled using a local excitatory-inhibitory (EI) circuit consisting of one E, one I population, and a mechanism controlling for adaptation (A) of the excitatory population. The implementation was guided by the EI network model proposed by Jercog et al. [11], who showed that this model produces UP and DOWN states similar to the UP and DOWN states that were observed in recordings from mice during anaesthesia [13].

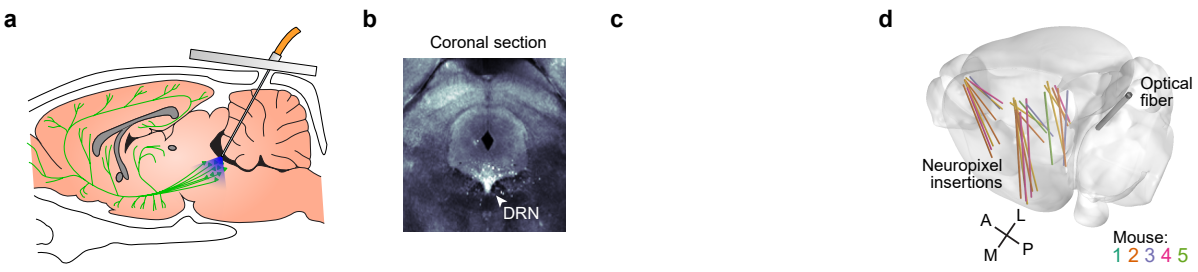


Figure 1: Acute Neuropixel recordings while optogenetically stimulating serotonin release. (a) Saggital cartoon drawing of serotonergic neurons in the DRN projecting throughout the brain (green). An optical fiber is positioned directly above the DRN through the cerebellum through which the 5-HT neurons can be activated with blue light. (b) A coronal histological section of the DRN showing expression of channelrhodopsin. (c) A coronal histological section showing the fluorescent tract of a Di-I coated Neuropixel insertion. (d) Neuropixel insertion trajectories, aligned to the Allen Brain reference atlas, across all seven mice. Typically seven insertions were made per mouse, color of each insertion indicates mouse identity.

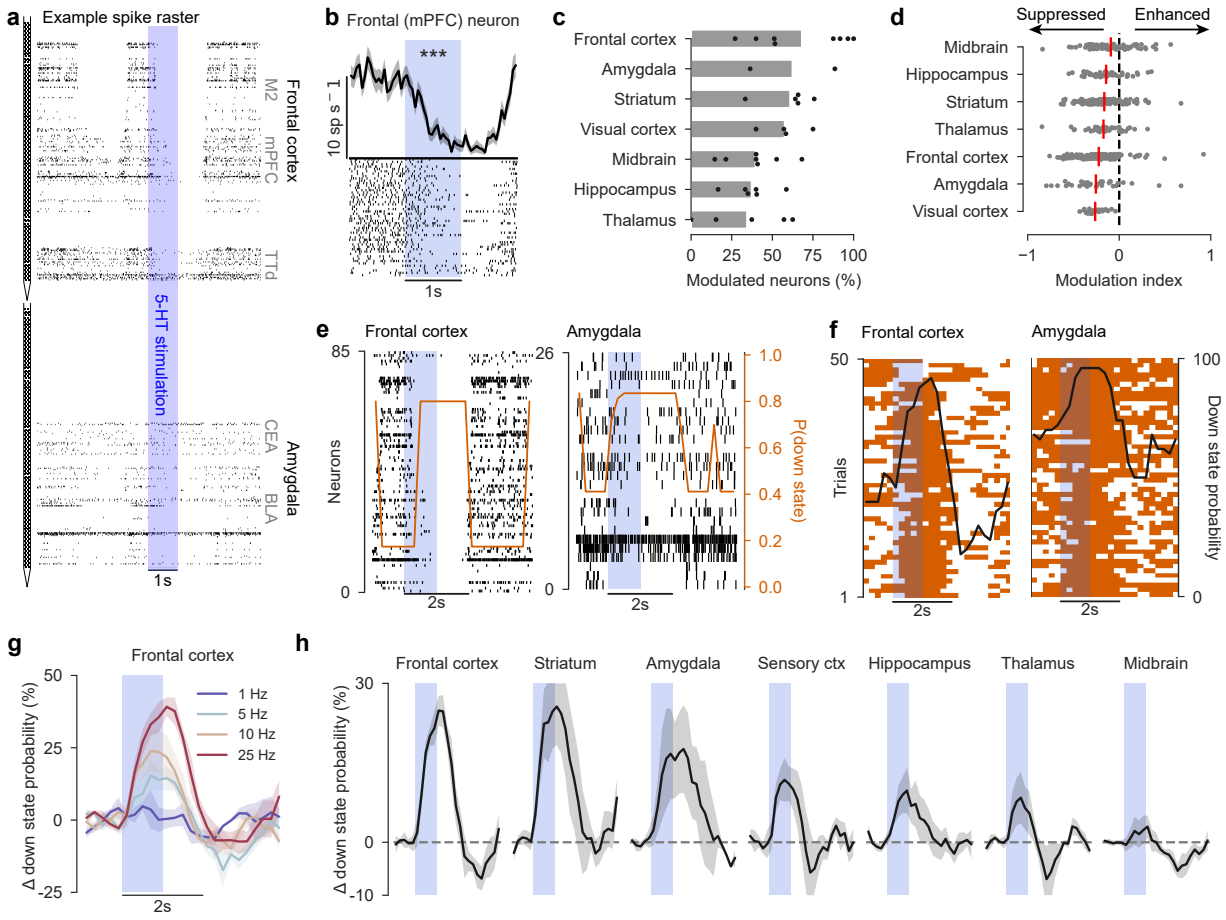


Figure 2: Serotonin induces a DOWN state under anesthesia. (a) An example spike raster plot of 10s of spiking activity recorded on two Neuropixel probes simultaneously. Recorded brain regions are indicated on the right of the raster at their approximate depths. The time of the optogenetic 5-HT stimulation is indicated by the blue bar (25 Hz, 10 ms pulses). (b) Peri-stimulus time histogram of an example neuron from the frontal cortex. The bottom shows the spike raster in which each row is a repetition of 1s 25 Hz stimulation. The top shows the mean spiking rate binned in 50 ms bins with a 25 ms Gaussian smoothing kernel. *** $p < 0.001$, ZETA-test. (c) Percentage of significantly modulated neurons (ZETA-test, $p < 0.05$) per brain region, calculated per recording session. Black dots are sessions, grey bars means over sessions. (d) Modulation index per neuron, negative values indicate the neuron is inhibited by 5-HT and positive values indicate excitation. Each grey dot is a single neuron, vertical red bars are the mean over neurons per brain region. (e)

1 Replication of mouse brain dynamics in an anaesthetic state

1.0.1 UP and DOWN states

The model was able to reproduce the slow-wave oscillations that were observed during anaesthesia in previous experiments. Figure **??** shows the results of a simulated session using the connected network model. Certain areas showed periods of high (UP) and low (DOWN) activity in the firing rate of the E and I populations. The frequency of the UP and DOWN states cycle was around 0.5-1Hz, which simulates the frequency of the UP-DOWN dynamics observed in the experimental data (Meijer et al., in preparation). The connections between the 14 areas on the figure were formed with a coupling strength of $G = 2$, to ensure that the UP-DOWN dynamics were synchronous across regions (more about synchrony in section 1.0.3 and Figure **??**). The slow-wave oscillations occurred without any serotonin interference, as this simulation did not include the stimulation by serotonin.

1.0.2 Heterogeneity in the global circuit

In the electrophysiological recordings it was observed that the UP-DOWN dynamics differed across brain regions. The hippocampus and midbrain areas clearly showed no state activity patterns, whereas

cortical areas and the striatum clearly showed UP and DOWN states. The amygdala, thalamus and piriform showed less clear UP and DOWN states. Adjusting the theta and beta parameter allowed for modification of the frequency of the UP-DOWN dynamics. This enabled introducing heterogeneity in the global network model making the simulation as close as possible to the experimental data. For example, in Figure **??** the hippocampus only showed an UP state for the entire duration of the model simulation. This replicates the findings from the electrophysiological data and could be reached by setting the adaptation value to $\beta = 0$ and $\theta_E = 5$.

1.0.3 Synchrony in the connected global network

The UP-DOWN dynamics that were visible during anaesthesia occurred in a synchronous rhythmic pattern. To evaluate the amount of synchrony in the network model, a measure proposed by Brunel and Hansel [14] was applied (see Section **??** for methodological details). Running the model simulation using different coupling strength values tests if the model follows this mechanism. Increasing the coupling strength increased the synchrony of the firing rates across the regions of the network. Since some of the areas did not show any UP and DOWN states, full synchrony ($\chi = 1$) is not possible to reach. If not otherwise specified, $G = 2$ was used in the following stimulation, corresponding to a value of $\chi = 0.55$ enabling enough synchrony in the system without entirely eliminating region-specific activity patterns. The effect of different coupling strengths on the effect of serotonin is evaluated in section 1.2.

In summary, the previous simulations using the global circuit model showed that the implemented model is capable of simulating brain activity during anaesthesia, which makes it a good tool to investigate serotonin's effects on UP-DOWN dynamics in a large-scale network.

1.1 Replicating serotonin's effect on UP-DOWN dynamics

Optogenetic stimulating of the DRN in mice during anaesthesia led to three main observations. In certain areas serotonin-induced 1) a DOWN state (strong effect in the frontal cortex, amygdala and striatum, weaker effect in sensory cortex, hippocampus, and thalamus), 2) a rebound UP state post-stimulation, and 3) a delayed onset of the response to the serotonin stimulation.

In the proposed large-scale model, serotonin stimulation was modelled by targeting the E population with an inhibitory current. Here, a serotonin stimulation of strength $S = 40$ was applied. In Figure **??**, the effect of serotonin in the model and in the experimental data are compared (the top panel represents the model results, and the lower panel represents the results from electrophysiological recordings).

It was clearly visible that during optogenetic stimulation of the DRN in mice (area highlighted in light blue), the probability of a DOWN state increased between 25-50% in the frontal cortex, amygdala and striatum, compared to the baseline activity, in which no stimulation was applied. The sensory cortex, hippocampus and thalamus showed an induced DOWN state too, though slightly less than the previously named areas (10-12%). The midbrain seemed not to be affected by the serotonin modulation. With simulations using the proposed model very similar effects can be observed. In the frontal cortex, striatum, sensory cortex, and thalamus the probability increased by about 50% in comparison with the baseline, and the amygdala showed an increase of even around 90%. Note that the DOWN state probability in the striatum was not significantly different from zero (see **??**), whereas this was the case in the other named regions. The midbrain shows a slight increase (significantly different from the baseline mean) of the DOWN state probability, different to the recordings in mice, where the midbrain was not affected by the stimulation. We observed that in the model the hippocampus was not modulated by serotonin, despite the observed 10% increased probability of a DOWN state in the mouse brain.

Some regions indicated the return of an UP state after the induced DOWN state in the mouse data, which are the frontal cortex, sensory cortex (though very marginal), and thalamus. In the model it is striking that the rebound UP state were slightly stronger expressed than in the mouse data; for example, the frontal cortex, sensory cortex and thalamus – overlapping with the affected regions in the experimental data -, and additionally in the striatum showed a rebound UP state. In the model,

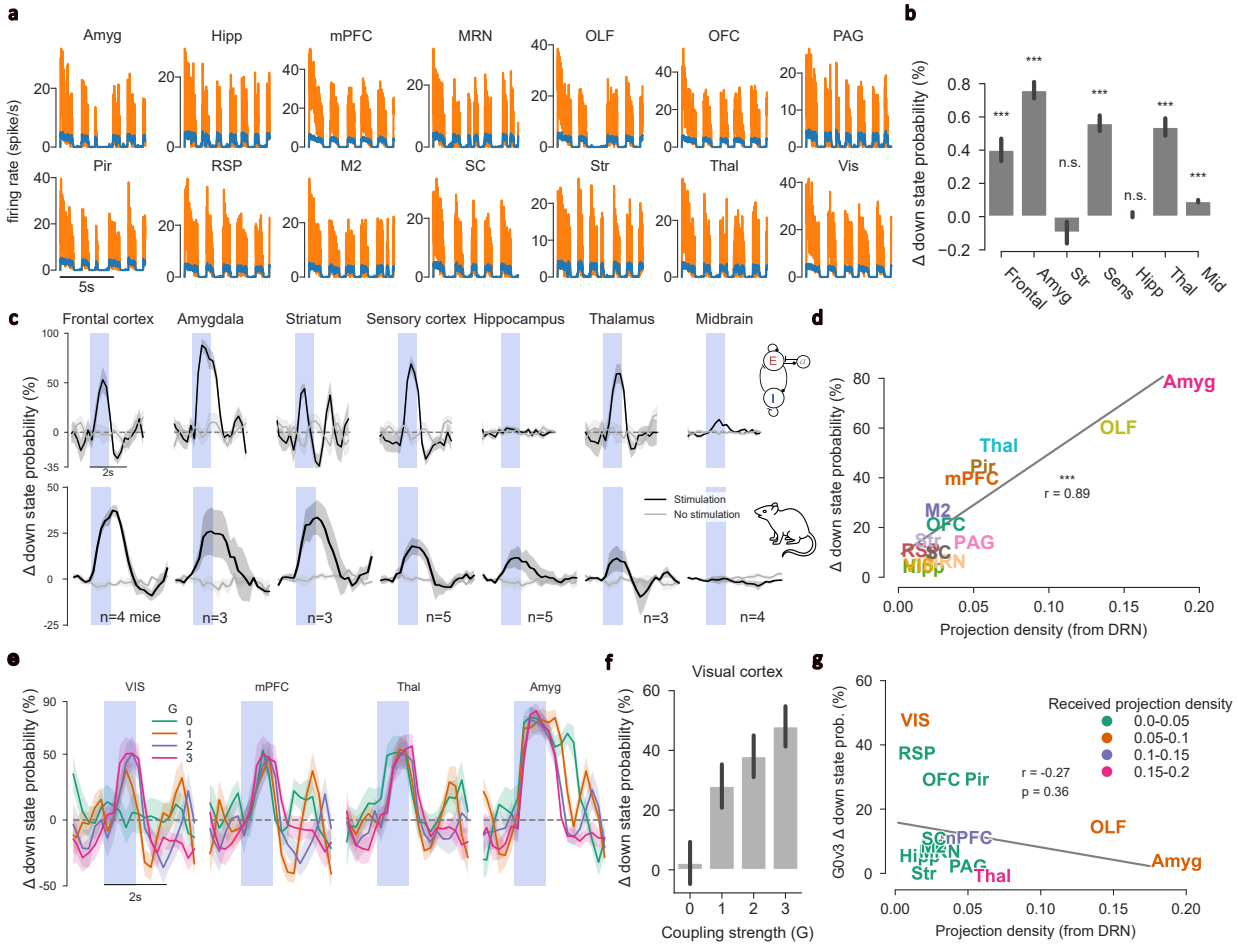


Figure 3: Simulation of UP-DOWN states, effect of serotonin and coupling strength. (a) Replication of UP and DOWN state dynamics. Without any serotonin stimulation, the activity in the brain regions UP and DOWN states. (b) Serotonin stimulation induces a DOWN state in several regions. Probability values between 0.5-1 seconds after the serotonin stimulation onset are included. (c) Direct comparison of the simulated and recorded DOWN state probability during serotonin stimulation. (d) In an unconnected network (G=0), the DRN projection density determines the strength of the effect of the modulation by serotonin. (e) Effect of coupling strength, demonstrated in example regions. The differently coloured graphs display different coupling strengths G from $0-3$. (f) The serotonin effect was elevated in the visual cortex for increased coupling strength (p¡0.05). (g) The difference between the Δ DOWN state probability for a coupling strength G=0 and G=3 is displayed for every region. The colour codes describe the average projection densities that each area receives from all other areas, indicating how strongly the areas are affected by the changes in coupling strength.

the rebound UP state was induced much faster than in the experimental condition.

What stands out is that the striatum showed strong fluctuations in the model simulations, in comparison to the other regions and in comparison to the measurements in mice. In all recorded areas displaying a modulatory effect, there was a delay of 0.5 seconds in inducing a DOWN state. This delay was simulated by implementing the time course of the serotonergic response. Overall, the main effect of serotonin that was observed in the electrophysiological recordings of mice could be replicated with minimal exceptions.

1.2 Effects of coupling strength

In this section, the potential underlying mechanisms behind serotonergic modulation will be addressed. One aspect to consider is the connectivity between the regions in the network, which are crucial for inter-area communication. The coupling strength of those inter-area projections was modified by increasing the value of the parameter G that defines the weight of the connectivity matrix W. If the effect of serotonin on the UP and DOWN dynamics acts in a direct way, inter-area coupling would not be required to replicate the induced DOWN state. Though, if the successful replication of the effect

requires inter-area coupling, serotonin acts indirectly. This mechanism might vary across different areas. We applied serotonin stimulation and systematically increased the coupling strength G, to gain insights into the role of inter-area coupling in controlling the effect of serotonin stimulation.

In an unconnected network (G=0), the probability of a DOWN state (compared to baseline activity) is dependent on the strength of the DRN projections targeting each area (see Figure **??** A). Regions to which the DRN strongly connects, showed a strong DOWN state probability during serotonin stimulation. The amygdala for example receives very strong connections from the DRN and therefore showed the highest modulatory effect of all of the regions in an unconnected network. The visual cortex, on the other hand, receives fewer projections from the DRN and is therefore almost not affected by the simulated serotonin stimulation.

To investigate the effect of coupling strength, simulations with various coupling strengths were run (G of $0 - 3$). Figure **??** demonstrates the resulting traces of DOWN state probability (each compared to the respective baseline) for the different coupling strength values in the visual cortex, medial prefrontal cortex, thalamus and amygdala. Modifying the coupling strength did not have the same effect in all areas. Whereas in the visual cortex, an increased coupling strength increased the DOWN state probability, the amygdala showed no big difference in DOWN state probability during serotonin stimulation for all G parameters. Additionally, it is noticeable that for $G = 1$ in the visual cortex and the mPFC, the rebound UP state is distinctively pronounced. In an unconnected network, the visual cortex was only slightly affected by serotonin stimulation (see Figure **??** B). It was observed that with an increased coupling strength, the probability of a DOWN state increased in the visual cortex; it was around 50% higher in the case of strong inter-area connections compared to no coupling.

To uncover if this effect was similar for all of the regions, the difference between Δ DOWN state probabilities in an unconnected network and a strongly connected network were investigated (see Figure **??** C). It turns out that whereas some areas were strongly affected by the change in coupling strength, others were not. The difference (G=0 vs G=3) was high for the visual cortex (p¡0.05), but close to zero for the amygdala. Areas that did not show any UP-DOWN states were not affected by the increase in coupling strength either.

This could not be explained by the strength of DRN connectivity, as the negative correlation was non-significant (r=-0.27, p=0.36). The projection densities (obtained from the Allan Brain Atlas) describing the connection strength between the areas co-existing in this network vary (see legend in Figure **??** C) and might affect how each area reacts to the coupling strength. The visual cortex receives relatively rich projections from other areas in the network, compared to the midbrain, striatum, thalamus, hippocampus, secondary motor cortex, and the remaining sensory regions. The thalamus showed, despite strong incoming inter-area projections, no enhanced serotonin effect for an increased coupling strength.

Overall, the coupling strength affected serotonin's induced DOWN state in some but not all areas, which could not be explained by the strength of DRN projections. Neither was the density of inter-area projections of each area a determining factor. Therefore, it is interesting to examine the heterogeneous connectivity pattern of the regions in this network model.

1.3 Heterogeneous connectivity

The previous results showed that the area specific connectivity pattern seems to be relevant for achieving an effect in some areas (see above in Section 1.2). Using information about the projection densities between areas (from the Allen Brain Atlas) implemented the area specific connectivity in the model and improved its biological plausibility. To investigate the relevance and impact of this effort on the obtained results, the simulations using the 'realistic' connectivity values were compared to simulations using homogeneous connectivity values, in which the mean of the entire connectivity matrix was applied as a single weight for all inter-area projections.

Previous evidence showed that inter-area communication introduces synchrony within a network [15, 16]. This might also be the case of the UP-DOWN dynamics that occur in each area. For example, if an UP state is induced in one area, the elevated firing rate is propagated to the E population of the other areas, increasing the chance of a simultaneous transition into an UP state in the targeted areas.

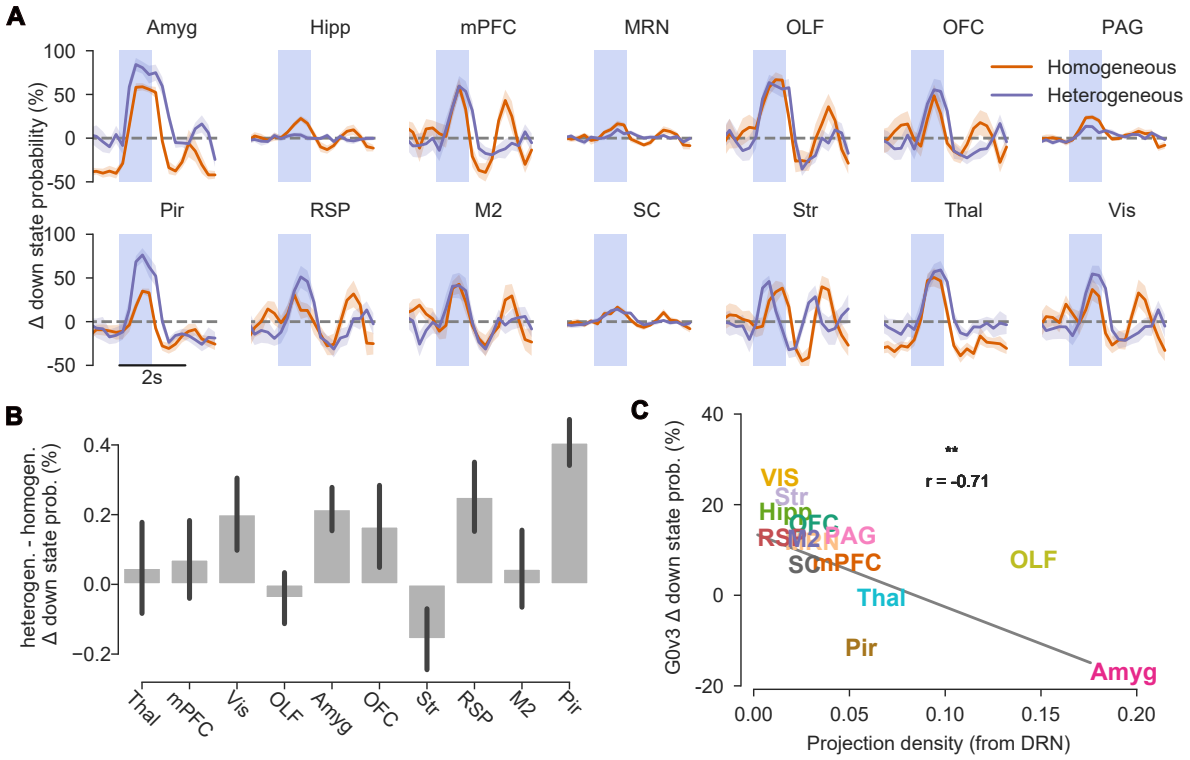


Figure 4: Heterogeneous vs. homogeneous connectivity. A) shows the trajectories of the DOWN state probability in homogeneous and heterogeneous connectivity compared to each of the baselines (coupling strength of $G = 2$). B) demonstrates the difference between homogeneous and heterogeneous connectivity seen in A). Regions that did not show any UP and DOWN areas were excluded. The regions on the x-axis are sorted by incoming projection strength (from all other areas) from high density to low density. In C), the difference in Δ DOWN state probability between $G = 0$ and $G = 3$ in the homogeneous condition is plotted and correlated with the DRN projection strength of each area. B) and C) include data in a time window of 0.5-1 seconds post-stimulation.

In the homogeneous condition, a DOWN state was induced upon serotonin stimulation in all areas (Figure 4 A). In some regions, the heterogeneous coupling mechanism strengthened the effect of serotonin in comparison to the homogeneous coupling, whereas in other regions the effect was reduced (see Figure 4 B). The thalamus for example, which receives strong projections from the other areas in the heterogeneous condition, did not show a great difference between the homogeneous and heterogeneous condition. The piriform cortex, which is an area that does not receive many projections from other areas, showed an increased effect of serotonin is higher in the heterogeneous condition. Another example is the secondary motor cortex, whose response did not change a lot in the homogeneous coupling condition even though it also receives few projections in the heterogeneous condition. These different results in the homogeneous and heterogeneous conditions might be created due to the specific areas from which both regions receive input, as well as the strength of the DRN projections that both areas received. In the heterogeneous condition, the piriform cortex received projections mainly from the olfactory cortex, which was strongly targeted by the DRN. This connectivity profile was overwritten by the homogeneous connectivity value.

In the homogeneous condition, there was a significant negative correlation (p¡0.01) found between the projection density originating from the DRN and the difference in serotonin-induced DOWN state of a coupling strength of $G = 0$ versus $G = 3$ (see Figure 4 C). The increased coupling strength ($G = 3$) had a higher influence on regions with low DRN projection densities. In some regions with a higher DRN connectivity, strong homogeneous coupling even reduced the DOWN state probability (e.g., in the amygdala).

In summary, heterogeneous inter-area coupling increased the efficacy by which of serotonin-induced DOWN states, compared to homogeneous coupling. Homogeneous coupling led to an evenly distributed effect across the regions.

2 The mechanism of serotonin stimulation

The mechanism used to simulate serotonin in the model plays a crucial role in replicating the effects observed in the experimental data. Here, the consequences of modifying the approach to serotonin stimulation within the model were investigated. Specifically, it was chosen to model serotonin stimulation as an inhibitory current targeting the E population ('$E-$'), as this increases the threshold for the E population to fire, making it likely to induce a transition from the UP to the DOWN state. It is also intuitive to consider the opposite operation of targeting the I population with an excitatory current ('$I+$'), as it may yield similar results. Figure **??** illustrates the outcomes of the simulation under this condition.

The graph reveals that, in many regions such as the Amygdala, cortical areas (OLF, OFC, M2, Vis, Pir, and RSP) and the striatum, a first spike of an induced DOWN state took place, followed by an UP state at around 1 second after stimulation onset. This is interesting since this is the opposite effect of what was observed when targeting the E population with an inhibitory current, as the main effect, the induced DOWN state could be observed around the same time as the UP state in this modified serotonin stimulation. Note, that the highest Δ DOWN state probability did not exceed 30%, which is very low in comparison to the probabilities found when targeting population E with an inhibitory current (see Figure **??**). In the somatosensory cortex as well as in the piriform cortex and thalamus, the initial DOWN state that occurred within the first 0.5 seconds after stimulation was longer and less 'spiky' that in the other regions named above. The serotonin stimulation acts gradually, therefore the current applied to I is rather weak at first. Nevertheless, the local circuit reacted to the weak current applied to the I population with an increased DOWN state probability. In turn, a stronger current had the opposite effect and let the system transition into an UP state.

Figure **??** B shows the direct comparison of the two distinct mechanisms ('$E-$' and '$I+$'). If both mechanisms were to induce the same effect, it was expected to observe a line centred around 0. Though, it is visible that the clear DOWN state in all regions except the hippocampus and the midbrain could not be replicated by the $I+$ mechanism.

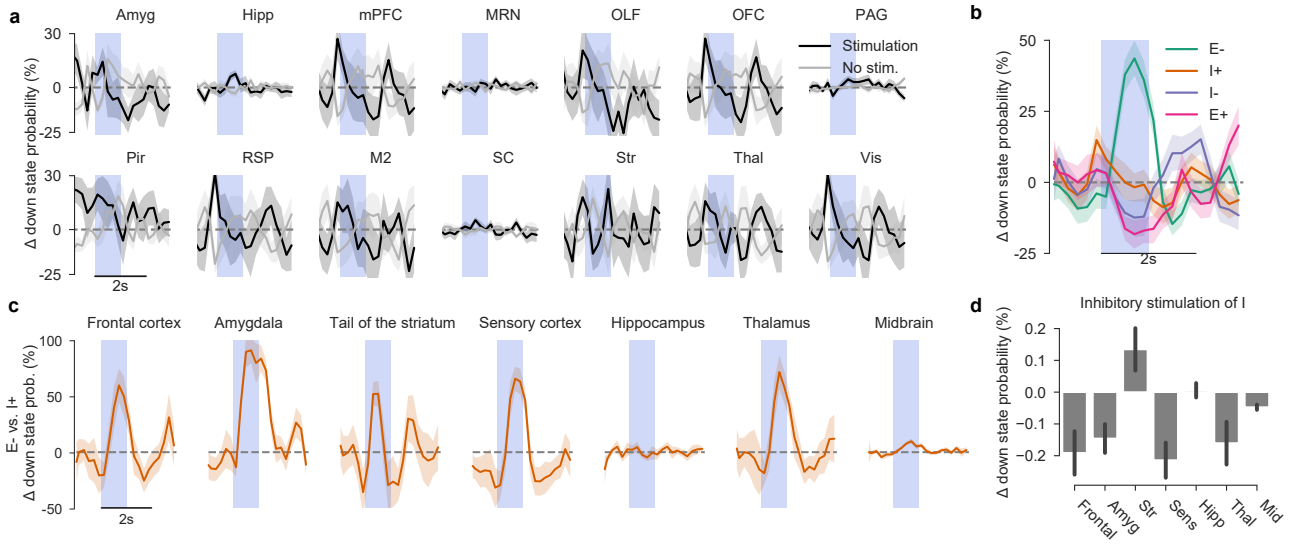


Figure 5: Target mechanisms of serotonin stimulation. (a) Modelling serotonin by targeting the I population with an excitatory current. Δ DOWN state probability in all simulated brain regions. (c) All four options of modelling serotonin stimulation, averaged over all regions in the network. The first letter in the legend description is the population targeted by serotonin, the sign after it indicates the charge direction of the current to that population. (c) Difference between targeting E with an excitatory current and targeting I with an inhibitory current. (d) shows a more detailed view of the 'I-' simulation.

To uncover the effects of other options for modelling serotonin stimulation, targeting the E population with an excitatory current ('$E+$') and targeting the I population with an inhibitory current were investigated ('$I-$'). The results of all four mechanisms ('$I+$', '$E-$', '$I-$', and '$E+$') are presented in Figure 5 A. It is evident that only the '$E-$' implementation successfully reproduced a stable induction of a DOWN state during serotonin stimulation, followed by a somewhat less probable rebound UP

state. As demonstrated previously, modelling the stimulation using the '$I+$' mechanism failed to replicate the desired effect. Notably, the '$E+$' and the '$I-$' mechanism led to the induction of an UP state with a probability of approximately 10-20% compared to the baseline (note that the inverse of the DOWN state probability equals the probability of an UP state). Figure 5 B shows that for the '$I-$' mechanism an UP state in frontal areas, amygdala, sensory areas and the thalamus was induced. Interestingly, a subtle induction of a DOWN state was observed in the striatum. This phenomenon could be attributed to the comparatively higher frequency of oscillations in the striatum relative to other areas.

The clear distinctions in the outcomes achieved by each mechanism highlight the complexity of the network model.

Discussion

Additional ideas for the discussion:

- De Filippo et al., (2021) find that targeting SST interneurons with 5HT2A receptor enables inducing the DOWN state. We find that targeting excitatory population is crucial for inducing a DOWN state. This might be because the model does not incorporate different receptor types. Also, they did record this with fMRI in combination with electrophysiological recordings - more long term effect? The different receptor type distribution of post-synaptic 5-HT receptors is more relevant for functional connectivity than the anatomical DRN projection density. Our model does not include different receptor types, therefore, there is a need for another modelling study investigating this
- In contrast to de Filippo (2021), Nekse (2016) describes that there is a "lack of an observed increase in inhibitory cell firing relative to Up state termination" and that "synaptic inhibition is not an important factor for Up state termination".
- Grandjean (2019):
  - Even though the areas have different receptor distributions, all showed a negative effect
  - More negative in: medial prefrontal cortex (mPFC), amygdala, striatum, hippocampal formation, cortical subplate, and isocortex
  - Downside of MRI: indirect measurement, and temporal resolution is limited
- What could be the potential modulatory mechanism of serotonin be?
  - UP states are sharp wave ripples
  - During quite wakefulness, stimulation of serotonergic neurons in the DRN supresses hippocampal sharp wave ripples
  - Previously it was found that serotonin suppresses hippocampal sharp wave ripples. Therefore, one possible mechanism might be that serotonin induces a DOWN state by supressing sharp wave ripples during the UP state

Insights from electrophysiological recordings from anaesthetized mice during optogenetic stimulation of the DRN showed that serotonin stimulation induces a DOWN state, mainly in frontal and sensory cortices as well as in the amygdala and striatum. These findings were the basis for the implementation of a firing rate model that simulates the brain activity of the measured regions and aims to uncover the mechanisms of serotonin modulation on oscillatory neural activity during anaesthesia.

The model used in this project is based on a previously proposed model [11] and extends it by connecting multiple neural curcuits to one comprehensive brain network to investigate the mechanisms of serotonin on UP and DOWN states in anesthesia, which has not been done before. Most studies investigating the serotonin's modulation of oscillations focus on a specific brain region (see [9, 17]). This project implements an approach that investigates serotonin's effect on a large scale and therefore provides exciting insights into its mechanism in the entire brain.

3 Replication of serotonin's effect

The simulations performed by the proposed model demonstrate that a DOWN state was induced in the frontal cortex, amygdala, sensory cortex, thalamus, and midbrain, mostly overlapping with the effect observed in the electrophysiological recordings. Notably, no DOWN state could be induced in the hippocampus. This might be due to the fact that the hippocampus is targeted by very few serotonergic projections according to the Allen Brain Atlas. Additionally, it does not receive strong connections from any of the other areas, which could spread the effect of serotonin. In the real brain, there are many more regions that might act as a relay station for information reaching the hippocampus, but since the model only comprises selected areas, this information transfer cannot be taken into account. Another reason might be the parameter setting of the model, determining the emergence UP and DOWN states in each area. It is difficult to determine these dynamics from the electrophysiological data, which is why the parameters are an estimation that might introduce differences between the model simulations and the recorded data.

4 Direct and indirect effects of serotonin

Without coupling between the areas, the effect of serotonin depends on the DRN projections targeting each area (see Figure **??** A), which is a direct effect induced by serotonergic neurons targeting each area. Introducing connections between areas and increasing this coupling strength results in an increase in the effect of serotonin in certain brain areas, for example in the visual cortex (see Figure **??** B). This indirect effect could not be observed in all areas, but which areas do show it does not depend on their initial serotonin effect, and also the density of incoming projections is not decisive (see Figure **??** C). The indirect effect of serotonin rather depends on the area specific connectivity pattern. The visual cortex, for example, does not receive strong projections from the DRN but receives many projections from other areas (such as the thalamus and the mPFC) that are strongly targeted by the DRN. For a high coupling strength (e.g., $G = 2$), the effect of serotonin can be communicated via the long-range projections, inducing a stronger DOWN state probability in the visual cortex upon serotonin stimulation compared to the effect observed when no coupling between areas exists.

The amygdala receives around the same density of projections from other areas (non-serotonergic) compared to the visual cortex, and still does not show an indirect effect with higher coupling values. The absence of this increased effect in the amygdala may be attributed to its connectivity profile. The amygdala receives robust projections from the DRN, rendering it significantly affected by serotonin stimulation even in the absence of long-range projections between areas. Areas showing a moderate DOWN state during serotonin stimulation provide relatively weak input to the amygdala, and therefore their effect does not substantially influence serotonin's modulation in this region. The thalamus serves as another example; it receives strong incoming projections from other areas, which would suggest the possibility of an indirect serotonin effect. However, no increased DOWN state probability is observed with increased coupling strength. Looking at the connectivity matrix, the visual cortex, which is minimally affected by serotonin release, is the area that targets the thalamus the strongest in this model. The visual cortex and other weak serotonergic target regions are seemingly not able to amplify serotonin's effect in the thalamus.

It is an important observation that increased coupling strength increases the synchrony in the system. As the weight between the areas strengthens, areas with a strong influence on others communicate their UP and DOWN state transitions to other regions, leading to simultaneous alternation between UP and DOWN states. Additionally, due to the increase in synchrony, the areas communicate induced DOWN states to other areas, amplifying serotonin's effect in certain brain regions. This is in line with previous findings from experiments as well as computational simulations, showing that inter-area communication gives rise to synchrony in the network and communicates incoming currents [15, 18]. Synchrony is the basis for many cognitive functions like memory consolidation, learning, and perception. Evidence suggests that serotonin influences synchrony in a network [17]. Though, in this project, serotonin did not lead to desynchronisation of the network but communicated the effect of serotonin to other areas.

In conclusion, these findings suggest that the "indirect effect" of serotonin is facilitated by the

global synchrony in the network and depends on the specific connectivity profile of each region.

4.1 Realistic connectivity

Evaluating the influence of heterogeneous inter-area coupling on the effect of serotonin on UP and DOWN states, the connectivity values in the connectivity matrix were replaced with the mean value of the projection densities stored in the matrix.

Under homogeneous coupling, the impact of serotonin is diminished in multiple regions compared to when the connectivity matrix from the Allen Brain atlas is used. In some regions to be closer to the experimental data, for example in the amygdala, the effect is not as strongly pronounced as compared to in the model simulations. However, it is important to note that direct comparisons are complicated since the model does not include all the regions and connections present in the brain. Nevertheless, these findings provide insights into how heterogeneous coupling affects the distribution of serotonin's effect. Notably, regions with fewer projections do not display an enhanced serotonin effect in the heterogeneous condition compared to the homogeneous condition. On the other hand, stronger connected regions show an increased serotonin effect in the heterogeneous condition (e.g., the piriform cortex). However, this does not hold for all regions; for example, the secondary motor cortex, which despite being strongly targeted by other regions is not strongly influenced by the coupling method. These observations indicate that the individual projection profile of a region plays a role in determining the effect, similar to the finding in the investigations regarding the coupling strength.

In summary, heterogeneous coupling enhances the serotonin effect, while homogeneous coupling leads to more uniform DOWN state probability trajectories with reduced region-specific characteristics.

4.2 Only inhibiting excitatory populations induces DOWN states

The way how serotonin is simulated in the model is a crucial step in reproducing the effects that were found in the experimental data. It is known from Jercog et al. [11], that increasing the threshold of the E population induces the transition to a DOWN state. Adding an inhibitory current to the E population has the same effect since the E population will require more positive input to counteract the inhibitory input to reach the threshold θ_E. Therefore it was expected that when simulating the serotonin stimulation by an inhibitory current, targeting the E population would induce a DOWN state in the system. Intuitively, the symmetric way of doing so - stimulating the I population with an excitatory current - should lead to a similar effect. Instead of the expected effect, exciting I lead to an UP state in all areas of the network. This phenomenon could potentially be attributed to the localised nature of inhibitory connectivity. Inhibitory neurons are only communicating with neurons in their local circuit, in contrast to excitatory neurons (e.g., pyramidal cells), which form excitatory connections beyond the immediate vicinity of their local area and thereby establish long-range projections with other regions in the network. Consequently, within a coupled network, the E population experiences a substantially greater influence compared to the I population, even when subjected to identical serotonin stimulation. To test if the long-range connections really are the determining factor in the failure of the '$I+$' mechanism, simulations without coupling should be analysed in future research.

In light of this counter-intuitive discovery, it is relevant to explore alternative approaches to modelling serotonin stimulation, to investigate if those might unexpectedly lead to similar effects. All possible implementation options are either targeting the E population with an inhibitory, or excitatory current or targeting the I population with an excitatory current.

The results show that only the '$E-$' mechanism can replicate the effect of inducing a DOWN state (see Figure 5 A). All three other implementations reveal a more or less strongly induced UP state upon stimulation.

4.2.1 Paradoxical effect

Inhibiting the inhibitory population unexpectedly induces a decrease in DOWN state probability in the I population as well as in the E population. The inhibition of the I population might not be sufficient to silence the area. On the contrary, it increases the chance that the population keeps

firing and staying or transitioning into an UP state. This might be because of the interconnectivity in the local circuit since the interneuron inhibition decreases the inhibition of the excitatory population. This paradoxical effect has been observed in computational neural circuits and tested in vivo [19]. In a biologically inspired circuit model, implemented by Garcia Del Molino et al. [20], a similar response has been found. The model consists of one pyramidal cell and three interneuron types. Externally inhibiting one of the interneuron populations paradoxically induced an excitatory effect in the same population. For the paradoxical response to emerge, a prerequisite is that the excitatory population possesses sufficient recurrent connections[19, 20]. Whether that is also the case in this finding, in future research the connectivity in the local circuit should be manipulated systematically. Furthermore, it should be noted that the local circuits within the network implemented in this project all possess identical levels of local connectivity strength, whereas, in reality, this connectivity is likely to vary.

Relating this finding to serotonin's mechanisms, this counter-intuitive response might potentially be realised by the inhibitory effect of the $5HT_{1A}$ receptors. These receptors have an inhibitory effect and are situated on interneurons, thereby leading to their inhibition.

This finding highlights the complexity that can be observed even in simplistic neural circuits, which may be used by serotonin. Serotonin is able to switch its effect in different brain states, such as distinct behavioural states [21]. This richness in modulating neural activity underlines the neurotransmitter's powerful role in regulating brain functions.

5 Methodological Limitations and Advantages

Since the model is built only in a limited biologically plausible way, it is not possible to draw conclusions about the role of different receptor types in the modulation of UP and DOWN states during anaesthesia. The fact that only inhibiting the E population could induce the observed DOWN state, might involve both, the inhibitory receptor 5-HT_{1A} and the excitatory receptor 5-HT_{2A}, as both receptors are located on both inhibitory interneurons as well as excitatory pyramidal neurons pre- and postsynaptically[1]. To uncover the interactive mechanism of the two receptor types, a more fine grained model including different receptor types must be built. For example, it is very likely that the mechanism differs across the regions. For example, because 5-HT_{1A} are rich in pyramidal cells axon hillock and soma in the prefrontal cortex, the dominantly inhibitory effect of serotonin in this region may be mostly mediated by 5-HT_{1A} [17], which might differ from how serotonin affects other regions. Implementing a network with single neurons and even different receptors would provide more insight into how exactly serotonin modulation works. Additionally, one possible optimisation in terms of biological plausibility would be to implement projections back to the DRN. Evidence shows that there are projections from many brain regions (e.g., from the prefrontal cortex) that regulate the release of serotonin [1]. Furthermore, the distance between the nodes in the real brain has not been taken into account in our model. Though, it has been shown that despite the distance, a phase lock of zero time lag between far away regions is possible [22].

Although there are many possibilities to extend the model implemented for this project, the model presented here is still well suited to address questions regarding the relevance of connectivity between network components and the functioning of serotonin on a larger scale. By avoiding an excessively specific implementation and keeping the number of variables low, the model offers greater interpretability. Additionally, the availability of state-of-the-art atlas data does not yet provide the necessary level of detail regarding the distribution of 5-HT receptor types among specific cell types. Moreover, developing more detailed models would significantly increase the computational complexity of simulations and analyses. Taking these factors into account, the proposed model strikes a balance that allows sufficient biological plausibility for the questions raised to allow meaningful investigation.

Material and methods

Animals

Seven SERT-Cre C57Bl/6 mice (four males and three females) were used for the purpose of this study. All experimental procedures were approved by the Portuguese Veterinary General Board.

Surgical procedures

Mice underwent two surgeries: (1) injection of the channelrhodopsin viral vector in the dorsal raphe nucleus and implantation of the optical fiber & headbar, and (2) the creation of three small craniotomies in the skull. Both these surgical procedures were performed as previously described in [8]. A minimum of two weeks was allowed for the virus to express and the mouse to recover after the first surgery. The second surgery was performed $\tilde{1}2$ hours preceding the first Neuropixel recording.

Neuropixel recordings

Acute anesthetized Neurpixel recordings were performed on four consecutive days. Mice were anesthetized with an induction dose of 3% isoflurane after which they were headfixed in the recording rig (hardware described in detail in [8]). A maintenance level of isoflurane of 0.8-1.5% was administered through a mask which was placed over the nose of the mouse. The tip of each Neuropixel probe was lowered in drop of Di-I with the micromanipulator prior to insertion. One or two Neuropixel probes, depending on the recording day, were inserted in the following coordinates:

Session	ML (μm)	AP (μm)	Depth (μm)	Angle (°)	Directionality
1	-1090	2000	3800	15	medial
1	-2184	-2000	6800	10	lateral
2	-946	-3300	3800	10	medial
2	-1227	2000	4590	10	lateral
3	-2240	-2000	4000	15	medial
3	-850	-3460	5270	10	lateral
4	-2250	-1800	5500	17	lateral

Throughout the recording session, the isoflurane level was calibrated such that the mouse would not wake up but UP and DOWN states were visible in the electrophysiological recordings. Once mice were stable under light anesthesia, serotonergic neurons in the dorsal raphe nucleus were optogenetically stimulated with 1s pulse trains of blue light (465 mm) administered through the optical fiber. The light power was 5 mW, as measured at the tip of the optical fiber. Light pulses were 10 ms long and with a frequency of 1, 5, 10, or 25 Hz. Different frequencies were administered in a randomized order and repeated each 50 times. The inter-stimulus interval was drawn from a truncated exponential distribution with a factor of 6s, a minimum of 5s, and a maximum of 15s. The whole stimulation protocol lasted around 30 minutes, during this time the body temperature of the mice was kept from falling using hand warmer packages (electrical heating pads produced large artifacts in the electrophysiological recordings). After the protocol, the Neuropixels were retracted, the skull covered, and the animal was placed in a cage positioned on a heating pad to wake up.

Processing of electrophysiological data

Pre-processing and spike sorting of raw electrophysiological data was performed with the fully automated pipeline developed by the International Brain Laboratory (IBL) [23]. We only included neurons that passed all the IBL neuron-level quality control criteria. Additionally we excluded neurons with firing rates < 0.1 spks/s over the whole recording.

Modulation index

For each neuron a modulation index was calculated which quantifies the strength and directionality of the modulatory effect of serotonin. A 500 ms baseline window was defined directly before stimulus onset, and a 500 ms stimulus window was defined at 300 - 800 ms after stimulus onset. The modulation index MI was defined as the area under the ROC curve $auROC$ between the spike counts in the baseline window and the stimulus window, which was scaled between -1 and 1 according to Equation 1.

$$MI = 2(auROC - 0.5) \tag{1}$$

Firing rate model

We implemented a bistable firing rate model, consisting of multiple local circuits which were all-to-all connected to form a global network. The local circuits represented the relevant brain areas from the experimental recordings. Serotonin stimulation was modelled by inducing a current to all regions, originating from the DRN. The connectivity was realistically modelled by using connectivity data from the Allen Brain Atlas which defined the connectivity between the regions in the model as well as the projection density from the DRN to the different regions.

The local circuit model was based on the implementation by Jercog et al. [11], including one excitatory population E and one inhibitory population I. It included an adaptation module a, simulating synaptic fatigue in pyramidal cells, as well as background noise which was introduced to the E and I population. Both populations projected to each other, as well as to themselves through self-coupling. The following equations describe the change of the populations E and I firing rate r (see Equation 2 and 3), as well as the strength of adaptation a (see Equation 4) over time, where τ_E, τ_I and τ_a are time constants:

$$\tau_E \frac{dr_E}{dt} = -r_E(t) + \varphi_E(J_{EE} r_E(t) - J_{EI} r_I(t) - a(t) + \sigma\xi_E(t)) \tag{2}$$

$$\tau_I \frac{dr_I}{dt} = -r_I(t) + \varphi_I(J_{IE} r_E(t) - J_{II} r_I(t) + \sigma\xi_I(t)) \tag{3}$$

$$\tau_a \frac{da}{dt} = -a(t) + \beta r_E(t) \tag{4}$$

The frequency of UP and DOWN state changes was defined by $\frac{1}{\tau_a}$, so the original value of $\tau_a = 500ms$ generated states in the frequency of around 2Hz. Analyses of the experimental data in the mouse revealed that the frequency of state change is 0.5-1Hz, therefore a value of $\tau_a = 1500ms$ was chosen and the other time constants were scaled accordingly, so that the final implementation yielded $\tau_E = 30ms$ and $\tau_I = 6ms$. The variable J defined the local coupling strength for the E and I population, which was $J_E I = 5$, $J_E E = 1$, $J_E E = 10$, and $J_E E = 0.5$ (the first letter indicating the target population). To model the noise of the system, which was introduced as a background noise to both populations, two Ornstein-Uhlenbeck processes were implemented, with a mean of zero and a standard deviation of $\sigma = 3.5$. The functions φ_E and φ_I were the transfer functions, which determined the reaction of the population to incoming currents, through self coupling, projections coming from other populations, or/and background noise. The equation is a threshold-linear function, with a threshold that defines the start of the linear part of the equation. If the input did not reach the threshold, the activation of the regarding population was set to zero.

$$\varphi_X(x) = g_X[x - \theta_X]_+ \qquad , X = \{E, I\} \tag{5}$$

The threshold that has to be reached until the population reacts to incoming input is described by θ. For brain regions that showed UP and DOWN states $\theta_E = 6$ and $\theta_I = 25$ were used as thresholds. In case the input reached the threshold level, the slope g determined the output. g controlled how sensitive the population reacts to input; a higher g will lead to a stronger output than a lower g. The slope g was fine-tuned for each brain region so that the average firing rate of the E and I population is slightly below the value measured in the mouse brain, since inter-area connectivity further increases the baseline firing rate. UP and DOWN state characteristics were also matched to the experimental data on a region-by-region level; some brain regions, like the cortex, showed strong UP-DOWN dynamics *in vivo* whereas other regions, like the midbrain, did not. To mimic this in the model, the adaptation strength β and the threshold of the E population θ_E were fine-tuned per brain region. For regions which showed strong UP-DOWN dynamics $\beta = 6$ and $\theta_E = 4$ was used, regions without UP-DOWN dynamics were modelled at $\beta = 1$ and $\theta_E = 5$, and intermediate regions at $\beta = 3$ and $\theta_E = 5$.

Modelling inter-area connectivity

Realistic connectivity was introduced to the model using data from the Mouse Connectivity Atlas (Allen Brain Institute). Brain regions which were targeted by the Neuropixel recordings were defined as regions as interest. For each region, we obtained the projection density values to all other regions. Results from all experiments per region were averaged. Since some experiments did not show any projection values, a threshold of a projection density of 0.005 was applied for the experiment to be included. As a next step, the obtained values of the fine grained areas were grouped together into higher level brain areas that each represent one component of the network (see Figure **??**). Besides the connectivity values between the areas within the network, the anatomical projections originating from the DRN were set according to tracing data from the Allen Brain Institute (see Figure **??**). The connectivity data was introduced to the global network model, by taking the projection density values W^{xy} (projection to area x from area y) and multiplying them with the coupling strength G between the inter-area projections. The input into population E of area x can therefore be defined as I_E^x from projections that originate from all other areas y (see Equation 6).

$$I_E^x = G \sum_y W^{xy} \tag{6}$$

Synchrony metric

To measure the synchrony between the brain regions in the model, we computed the parameter $\chi(N)$ proposed by Brunel [14] (see Equation 7 and 8). The idea behind the formula is to take the variance of each region and compare it to the variance of the entire network. N denotes the number of local areas in the network, and r is the I population's firing rate.

$$\sigma_r^2 = \langle [r(t)]^2 \rangle_t - [\langle r(t) \rangle_t]^2 \tag{7}$$

$$\chi(N) = \sqrt{\frac{\sigma_x^2}{\frac{1}{N}\sum_(^N i = 1)\sigma_{r_i}^2}} \tag{8}$$

Modelling 5-HT stimulation

The modulatory effect of serotonin stimulation was modelled by a current from the DRN projecting to the local circuits in the model. Similar to the connectivity matrix for the inter-area long-range projections, the projections of the DRN are weighted depending on the projection density. Considering that the DRN projection densities to all areas x are stored in a vector $\mathbf{D}$ the input current I_{DRN} is defined as the following:

$$I_{DRN}^x(t) = S\mathbf{D}_x u(t) \tag{9}$$

The temporal dynamics of the release of serotonin in the model was matched to the temporal profile of the modulatory effect of 5-HT on neural activity (data not shown). The trajectory of 5-HT modulation peaked after 0.5 seconds and decayed it returns to the original state after 3 seconds. The time course of this trajectory, here defined as $u(t)$, was multiplied with the parameter for serotonin stimulation strength S. The stimulation was given for 1 second, with an inter-stimulus interval similar to the experimental protocol. The effect of serotonin stimulation on the system was initially modelled by introducing an inhibitory current to the E population (subtracting I_{DRN} from r_E), this was later extended to other mechanisms.

Hidden Markov Model (HMM)

In both the electrophysiological and simulated data, UP and DOWN states were quantified by fitting an HMM using the `ssm` package [24]. In the case of the electrophysiological data, spike times were binned in 200 ms bins in seven second windows (2s pre, 5s post) surrounding 5-HT stimulation onsets.

An HMM with two states and Poisson observations was fit to the binned spike rates. In the case of the model, an HMM with two states and Gaussian observations was fit to the modelled firing rate time series of the I population. The state which showed the highest firing rate was determined to be the UP state. The output of the HMM was an inferred UP or DOWN state per time point and a corresponding posterior likelihood.

Data and code availability

Details here

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Conflict of interest disclosure

The author of this preprint declares to have no financial conflict of interest with the content of this article.

Appendix

UP and DOWN state dynamics

Several components of the model play a role to generate the UP-DOWN dynamics. Most importantly, the system requires bistability, meaning that the model has two stable states of activity, which are called stable attractor points. When one of those points is reached, the network stays at that rate of activity if there is no perturbing signal from the outside reaching the network. The first attractor point represents the DOWN state, in which no firing rate is observed. The second attractor point represents the UP state, with an average firing rate that depends on the setting of local connectivity parameters (self coupling and coupling between E and I). The bistability of the local circuit is reached by the coupling between E and I population and the way the transfer functions are implemented. To find the two attractor points of the system, refer to the nullclines of $r_E(t)$ and $r_I(t)$. Equation 5 and 10 describes $r_E(t)$ and $r_I(t)$, and when solving the equation with setting $r_E = 0$ and $r_I = 0$, it yields the nullcline of each population. Under specific conditions the nullclines intersect at two points, which are the two attractor points in which both firing rates stay the same.

$$r_E = g_E[J_E E r_E - J_E I r I - a - \theta_E]_+ \tag{10}$$

$$r_I = g_I[J_I E r_E - J_I I r I - \theta_I]_+ \tag{11}$$

Where the two nullclines intersect depends on the threshold θ of both populations, since it defines the start of the incline of the linear function. If $\theta_E < \theta_I$, the nullcline of r_E has time to incline linearly, whereas the nullcline of r_I starts rising from zero only later on (see Figure 6 A). The fact that $g_E < g_I$, allows the nullclines to intersect and therefore allowing the stable attractor point for the UP state arise (see Figure 7 A).

Adaptation and the fluctuating noise in both populations can initiate the change between the two attractor points of the system. The threshold of the excitatory population θ_E and the adaptation strength β are two parameters that determine the transitions between one state to the other. Figure 7 C) visualises the change of UP and DOWN state frequencies in a phase diagram of the parameters β and θ_E.

The circuit dynamics can be described with a limit cycle regime with oscillatory pattern of $\frac{1}{tau_A} Hz$. The originally defined time constants produce UP and DOWN activity with the frequency of 2 Hz. Since the electrophysiological recordings from the mouse brain showed that the UP and DOWN states

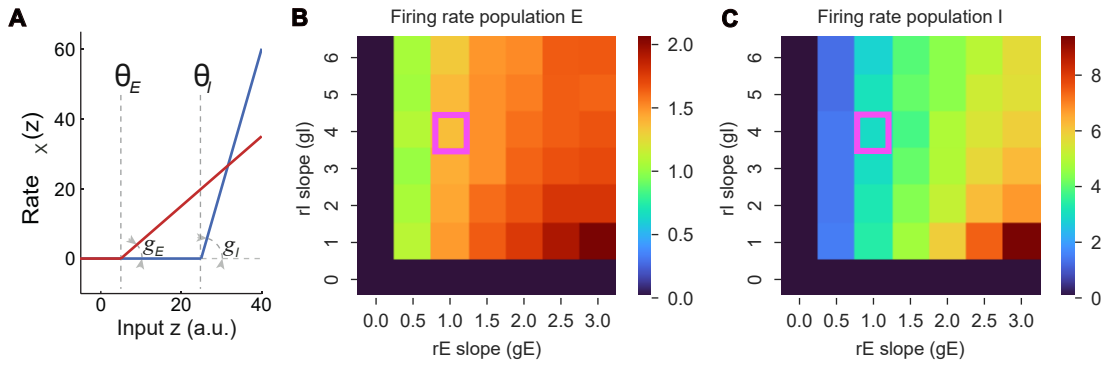


Figure 6: A) Nullclines of rE and rI (taken from Jercog et al. [11]). The effect of the nullclines' slope on the B) E and C) I populations' firing rate. The purple square indicates the default parameter setting.

have a frequency of 0.5-1 Hz, all time constants in the model were multiplied by 3, which leads to lower frequencies (see Figure 7 C).

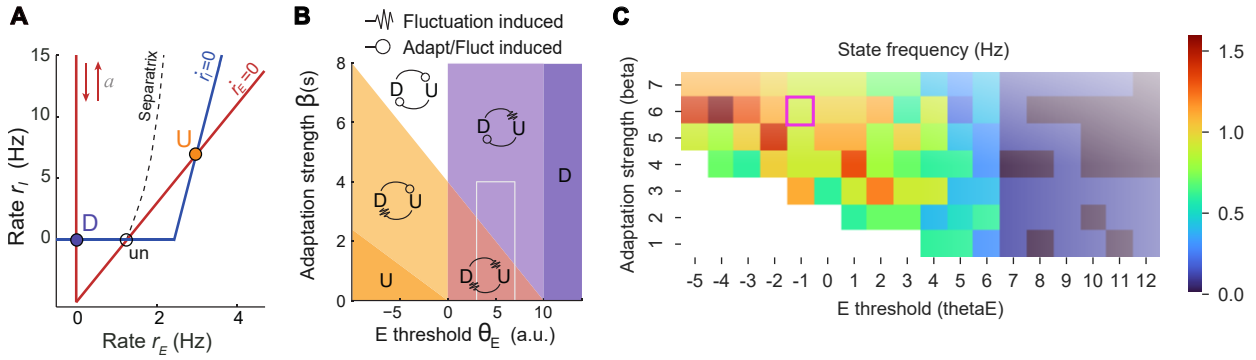


Figure 7: The role of the excitatory firing threshold and the adaptation strength in the UP-DOWN dynamics. A) Adapted from Jercog et al. [11], showing the two attractor points (U and D for UP and DOWN state) where the nullclines of rE and rI intersect. The current adaptation α value determines the position of the rE nullcline on the y-axis which can lead to the switch between the two attractor points. B) Taken from Jercog et al. [11], showing the phases in which the cuircuit switches between states with transitions induced by noise fluctuations or adaptation. C) Phase diagram showing the frequency of UP-DOWN states depending on the excitatory threshold and adaptation strength (using original time constant scaled by 3). Empty fields indicate absence of UP-DOWN states. The purple square indicates the setting for regions that show UP and DOWN states.

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